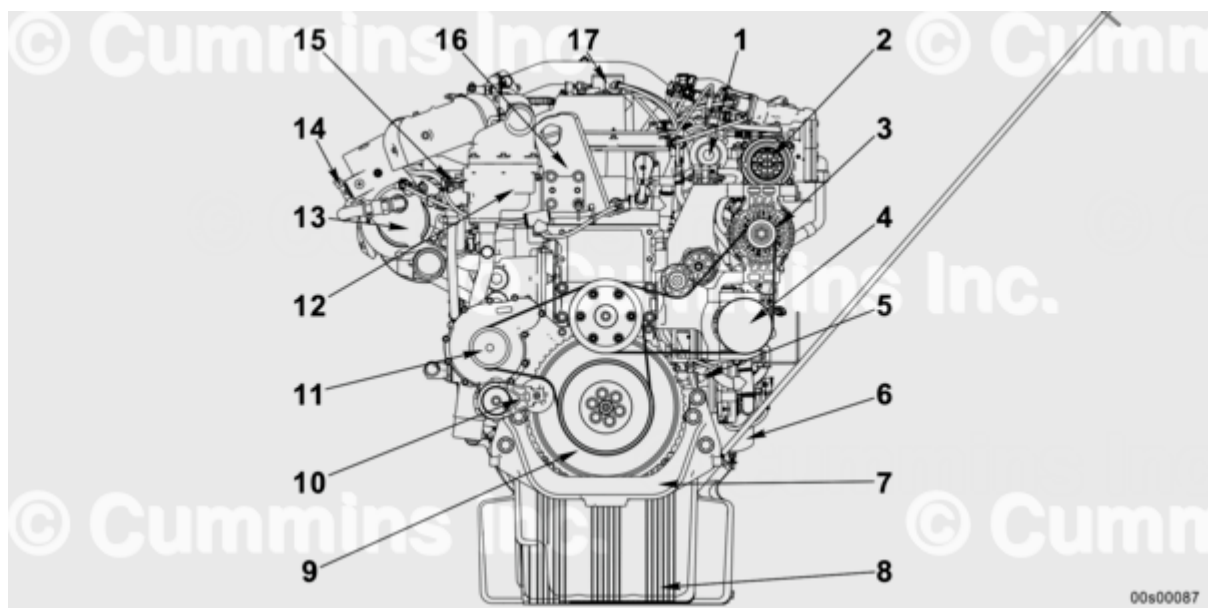


## Trainings ▾

### Engine Views

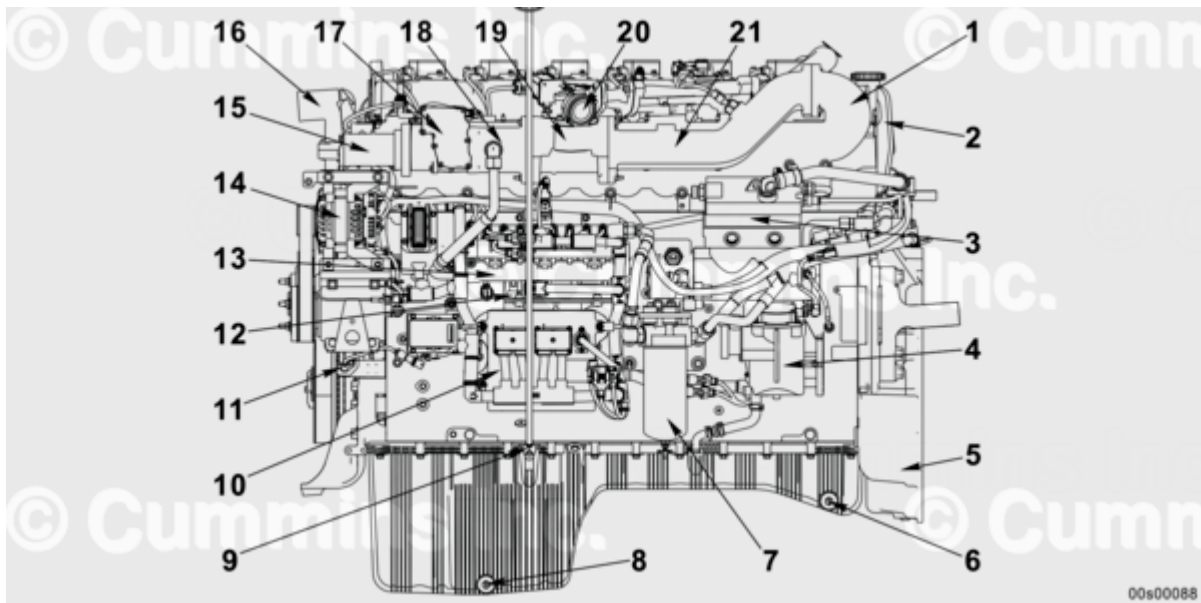
The following illustrations show the locations of the major external engine components, filters, and other service and maintenance points. Some external components will be at different locations for different engine models.



Front View

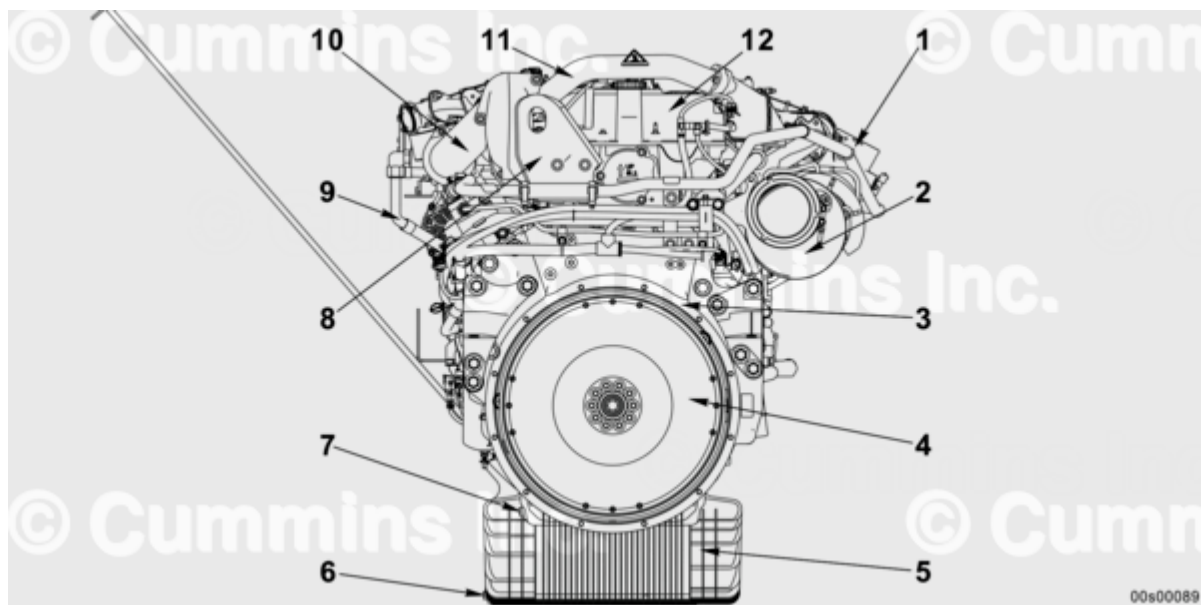
1. Intake manifold
2. Air intake connection
3. Alternator
4. Refrigerant compressor
5. Crankshaft position sensor
6. Fuel filter (spin-on type)
7. Engine support bracket, front
8. Lubricating oil pan
9. Vibration damper
10. Water pump belt tensioner
11. Water pump
12. Dual thermostat housing
13. Turbocharger compressor outlet connection
14. Turbocharger compressor intake humidity/pressure/temperature sensor
15. Engine coolant temperature sensor
16. Engine lifting bracket

17. Ignition coil.



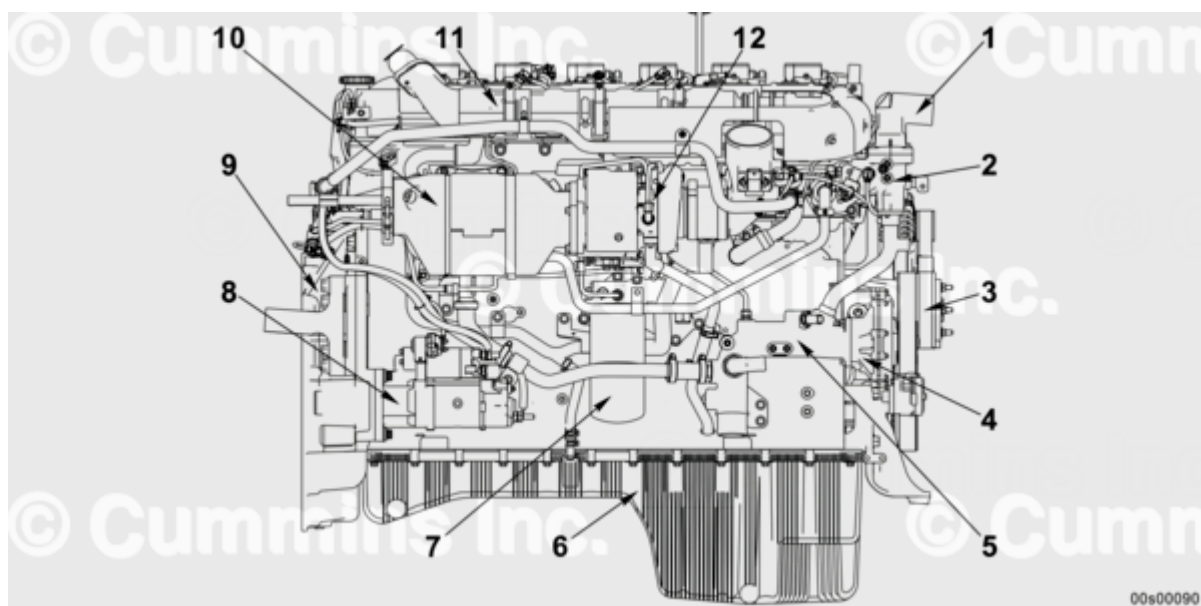
Left Side View

1. Intake manifold
2. Engine lifting bracket
3. Air compressor
4. Crankcase breather
5. Flywheel housing
6. Lubricating oil pan drain plug
7. Fuel filter (spin-on type)
8. Lubricating oil pan drain plug
9. Lubricating oil dipstick
10. Engine Control Module (ECM)
11. Crankshaft position sensor
12. Injector and Ignition Control Module
13. Fuel metering valve
14. Charging system alternator
15. Air intake connection
16. Coolant outlet connection
17. Engine intake throttle actuator
18. Air/fuel mixer
19. Exhaust Gas Recirculation (EGR) mixer
20. EGR valve
21. Air intake connection.



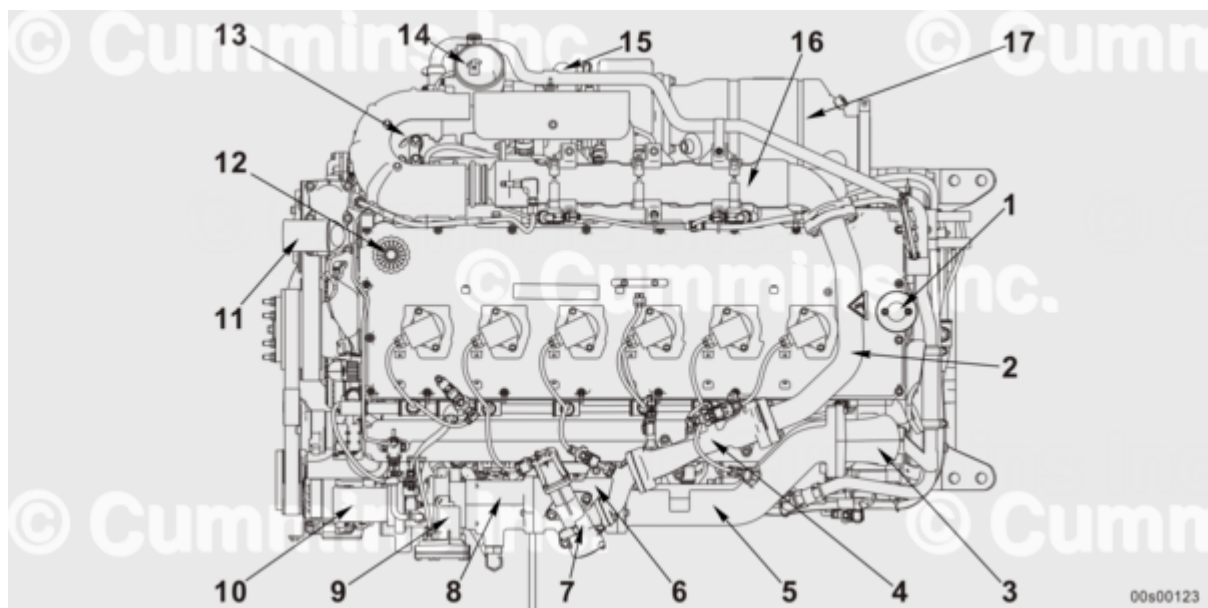
Rear View

1. Air inlet connection
2. Three-way catalyst
3. Flywheel housing
4. Flywheel
5. Lubricating oil pan
6. Lubricating oil pan drain plug
7. Lubricating oil pan drain plug
8. Engine lifting bracket
9. Fuel supply line
10. Air intake connection
11. EGR crossover tube
12. Rocker lever cover.



Right Side View

1. Coolant outlet connection
2. Coolant thermostat housing
3. Fan hub
4. Water pump
5. Water pump inlet connection
6. Lubricating oil pan
7. Lubricating oil filter
8. Starting motor
9. Flywheel housing
10. Three-way catalyst
11. EGR cooler
12. Turbocharger.



Top View

1. Oil fill cap or pressure relief valve
2. EGR crossover tube
3. Intake manifold
4. EGR venturi tube
5. Air intake connection
6. EGR mixer
7. EGR valve
8. Air/fuel mixer
9. Engine intake throttle actuator
10. Air intake connection
11. Water inlet connection
12. Oil fill cap
13. EGR cooler connection
14. Turbocharger compressor outlet connection
15. Turbocharger
16. EGR cooler

17. Three-way catalyst.

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**Last Modified: 21-Dec-2023**

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